Constant Maturity Swap Rates: Two-Factor Hull–White, Quasi-Gaussian and SABR-LMM Models for the Pricing of a CMS Steepener

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Abstract

[TODO] Write last. One paragraph: what a CMS rate is, why its expectation under the payment measure differs from the forward swap rate (convexity adjustment), why a steepener needs a model with imperfect correlation between the 2y and 10y swap rates, the three model families studied (G2++, multi-factor Cheyette with local/stochastic volatility, SABR-LMM), the data used (`EUR_SWAP_RATES`) and the headline numbers (convexity adjustments, steepener prices across models, calibration errors). This paper is linked to my GitHub, which can be found on my portfolio[1], and all scripts are available in the same repository.

[1] https://portfolio-eliasbenguedda.fr

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1 Introduction

1.1 Motivation

[**TODO**] CMS-linked products (CMS swaps, CMS caps/floors, CMS spread options, steepener notes, range accruals on CMS spreads) are among the most traded rate exotics in EUR. Their payoff depends on a swap rate observed in the future and paid at a date unrelated to the swap's natural annuity: a linear payoff in the rate is not linear in the model.

1.2 The two modelling questions

1. **Convexity**: the value of a single CMS rate depends on the smile of swaptions of the same tenor (static replication).
2. **Correlation**: the value of a CMS spread (steepener) depends on the joint dynamics of two swap rates, so it needs a model with at least two factors.

[**TODO**] Explain why a one-factor model (the HW1F model of the companion report) gives near-perfect correlation of the 2y and 10y rates and therefore misprices spread options.

1.3 Organization of the paper

[**TODO**] Write once the structure is final.

2 Preliminaries and Notation

2.1 Discount curve, forward rates and measures

[TODO] $P(t,T)$, bank account, $\mathbb{Q}$, T-forward measure $\mathbb{Q}^T$, change of numeraire. Multi-curve remark: €STR discounting vs EURIBOR 6M projection, and which one is used here (see Section 4).

2.2 Par swap rate and annuity

For a swap starting at T_0 with fixed dates $T_1 < \cdots < T_n$ and accrual fractions τ_i,

$$A(t) = \sum_{i=1}^{n} \tau_i P(t,T_i), \qquad S(t) = S(t;T_0,T_n) = \frac{P(t,T_0) - P(t,T_n)}{A(t)}. \tag{1}$$

Under the annuity (swap) measure $\mathbb{Q}^A$, $S(t)$ is a martingale.

2.3 Swaptions and the Bachelier formula

[TODO] Payer swaption $= A(0)\,\mathbb{E}^A[(S(T_0)-K)^+]$; Bachelier price with normal volatility σ_N; why normal vols are quoted in EUR (negative rates 2015–2021).

3 Constant Maturity Swap Rates

3.1 Definition

Definition 3.1 (CMS rate)**.** The n-year CMS rate fixing at T is the par swap rate $S_n(T) = S(T; T, T+n)$ of a spot-starting swap of tenor n observed at T. In EUR it is fixed daily by ICE Benchmark Administration (*ICE Swap Rate*) against EURIBOR 6M.

[TODO] Market conventions: fixing time, annual 30/360 fixed leg, 6M floating leg, fixing lag, payment in arrears/in advance.

3.2 CMS products

[TODO] CMS swap, CMS caplet/floorlet, CMS spread option, CMS steepener note. Give the steepener payoff used in this paper:

$$\text{Coupon}_k = \tau_k \, \min\Big(\max\big(L\,(S_{10}(T_k) - S_2(T_k)) + m,\ \text{floor}\big),\ \text{cap}\Big). \tag{2}$$

[TODO] Fix the term sheet: notional, maturity, leverage L, margin m, floor, cap, fixing in advance/in arrears.

3.3 The CMS convexity adjustment

A CMS coupon paid at T_p is valued under $\mathbb{Q}^{T_p}$, under which S is *not* a martingale:

$$\mathbb{E}^{T_p}[S(T)] = \frac{A(0)}{P(0,T_p)}\,\mathbb{E}^{A}\Big[S(T)\,\frac{P(T,T_p)}{A(T)}\Big] \neq S(0). \tag{3}$$

[TODO] Interpretation: the ratio $P(T,T_p)/A(T)$ is increasing in S, so the adjustment is positive and grows with volatility, maturity and tenor.

3.4 Static replication

[TODO] Write $P(T,T_p)/A(T) \approx G(S(T))$ (annuity mapping function), then use Carr–Madan to express a CMS caplet as a strip of payer swaptions:

$$\text{CMS caplet}(K) = G(K)\,\text{Swpt}(K) + \int_K^\infty \big[(S-K)G(S)\big]''\,\text{Swpt}(S)\,\mathrm{d}S. \tag{4}$$

[TODO] Annuity mappings: Hagan's standard model, the linear terminal swap rate (TSR) model. Requires a smile extrapolation (SABR) for large strikes.

3.5 Closed-form approximations

[TODO] Hagan's approximation with flat Black/Bachelier vol; timing adjustment when $T_p \neq T$. Numerical comparison with replication in Section 8.

3.6 Why the spread needs a correlation model

[TODO] The convexity adjustment only depends on marginal smiles, but the spread option depends on the copula of (S_{10}, S_2). Bachelier spread formula with correlation ρ; sensitivity of the steepener to ρ.

4 Data

4.1 The `EUR_SWAP_RATES` dataset

The file `Data/EUR/EUR_SWAP_RATES.csv` contains daily EUR par swap rates, in percent, for the tenors 1y–12y, 15y, 20y, 25y, 30y, 35y, 40y, 45y and 50y, from 1 January 2000 to 18 August 2022 (8 266 rows). **[TODO]** Confirm the conventions (fixed leg annual 30/360 against EURIBOR 6M?) and the source. **[TODO]** Cleaning: drop weekend/holiday rows (e.g. 1–2 January 2000 duplicate the last fixing), handle missing tenors (11y and 35y–50y are empty in the early years), choose a pricing date.

4.2 What the dataset allows and what it does not

Remark 4.1 (Data limitation)**.** The dataset contains **swap rates only**. It gives:

- the initial EUR curve on any date (bootstrap of discount factors from par rates, single-curve, or with €STR discounting using `ESTR_Fixings.csv` after October 2019);
- the *historical* dynamics of the curve: principal components, realised normal volatilities of each CMS tenor, and the historical correlation of S_2 and S_{10} and of their changes.

It does **not** contain swaption or cap volatilities, so the risk-neutral volatility parameters (G2++ σ, η, ρ; Cheyette local/stochastic volatility; SABR α, β, ρ, ν and LMM correlations) cannot be calibrated to market prices from this dataset alone. They must be either (a) calibrated to *illustrative* swaption volatilities, as in the Hull–White one-factor report, or (b) estimated under the historical measure from the time series. Every number in the paper states which choice is used.

[TODO] Decide between (a), (b) or both. Option (b) is a legitimate approach for correlation (the parameter the market quotes least), less so for volatility levels.

4.3 Exploratory analysis

[TODO] Figures: history of S_2, S_{10} and $S_{10} - S_2$ since 2000; rolling realised normal vol; rolling correlation of daily changes; PCA of daily changes (level, slope, curvature) and explained variance. The PCA motivates the number of factors.

5 The Two-Factor Hull–White Model (G2++)

5.1 Definition

$$r(t) = x(t) + y(t) + \varphi(t), \qquad \begin{aligned} \mathrm{d}x &= -a\,x\,\mathrm{d}t + \sigma\,\mathrm{d}W_1, \\ \mathrm{d}y &= -b\,y\,\mathrm{d}t + \eta\,\mathrm{d}W_2, \end{aligned} \qquad \mathrm{d}\langle W_1, W_2\rangle_t = \rho\,\mathrm{d}t. \tag{5}$$

[**TODO**] Link with the HW2F parametrisation (Hull–White 1994) and with the PCA of Section 4.

5.2 Fit to the initial curve and bond prices

[**TODO**] $\varphi(t)$ in closed form; affine bond price $P(t,T) = \frac{P(0,T)}{P(0,t)}\exp(\mathcal{A}(t,T) - B_a(t,T)x(t) - B_b(t,T)y(t))$.

5.3 Instantaneous correlation of swap rates

[**TODO**] Derive the instantaneous correlation of two forward rates (or swap rates) as a function of a, b, ρ. Show that it can go well below 1, unlike HW1F, which is the reason to use the model for a steepener.

5.4 Swaption pricing

[**TODO**] Brigo–Mercurio one-dimensional integral; or swap-rate normal-vol approximation by freezing the weights.

5.5 CMS and CMS spread pricing

[**TODO**] Monte Carlo with exact simulation of $(x, y, \int r)$; $S_n(T)$ computed from the affine bond formula, so no inner simulation. Compare the convexity adjustment with the replication of Section 3.

5.6 Calibration

[**TODO**] Parameters to calibrate: $(a, b, \sigma, \eta, \rho)$. Calibration targets: swaptions on the 2y and 10y tenors (illustrative vols) and/or the historical correlation of $\Delta S_2, \Delta S_{10}$. Known issue: ρ close to -1 and degeneracy between factors.

5.7 Limitations

[**TODO**] Gaussian rates, no smile, so CMS convexity is driven by a flat normal vol; the correlation structure is rigid in time.

6 Multi-Factor Quasi-Gaussian (Cheyette) Models

6.1 From HJM to a Markovian representation

[TODO] HJM with separable volatility $\sigma_f(t,T) = h(T)^\top g(t,\cdot)$; Ritchken–Sankarasubramanian / Cheyette condition; state variables $x(t) \in \mathbb{R}^d$, $Y(t) \in \mathbb{R}^{d\times d}$.

6.2 The d-factor quasi-Gaussian model

With a diagonal mean-reversion matrix κ and $G(t,T) = \int_t^T e^{-\kappa(u-t)}\mathbf{1}\,\mathrm{d}u$,

$$\mathrm{d}x(t) = \big(Y(t)\mathbf{1} - \kappa\, x(t)\big)\,\mathrm{d}t + \sigma_r(t,x)^\top \mathrm{d}W(t), \tag{6}$$

$$\mathrm{d}Y(t) = \big(\sigma_r^\top \sigma_r - \kappa Y - Y\kappa\big)\,\mathrm{d}t, \tag{7}$$

$$P(t,T) = \frac{P(0,T)}{P(0,t)} \exp\Big(-G(t,T)^\top x(t) - \tfrac{1}{2} G(t,T)^\top Y(t)\, G(t,T)\Big). \tag{8}$$

[TODO] Check the formulas against Andersen–Piterbarg, Vol. II, Ch. 13.

6.3 Local volatility specification

[TODO] Linear local volatility in the swap rate (displaced diffusion): $\sigma_r(t,x) = \lambda(t)(\alpha + \beta\,(S(t,x) - S(0)))$; controls the skew.

6.4 Stochastic volatility specification

[TODO] Multiply by $\sqrt{z(t)}$ with $\mathrm{d}z = \theta(1-z)\mathrm{d}t + \eta\sqrt{z}\,\mathrm{d}Z$; controls the curvature of the smile.

6.5 Swaption pricing by Markovian projection

[TODO] Swap rate dynamics in the model, freezing the swap-rate gradient, projection onto a displaced Heston model for S (Piterbarg); this gives the fast calibration. Not implemented: the repository calibrates the smile parameters by Monte Carlo with common random numbers, on top of the G2++ Gaussian core. State this and its cost.

6.6 Numerical methods

[TODO] Monte Carlo (Euler with Y integrated deterministically for the local-vol case) and PDE for $d \le 2$. CMS rates directly from the bond formula.

6.7 Calibration

[TODO] Per-expiry calibration of $\lambda(t)$ (level), β (skew), η (curvature); correlation between factors from the PCA or historical CMS correlation.

6.8 Limitations

[TODO] Approximate calibration, numerical cost, factor correlation still parametric.

7 The LIBOR Market Model and SABR-LMM

7.1 Forward rate dynamics

[TODO] Simply compounded forwards $F_i(t) = F(t; T_{i-1}, T_i)$ on the EURIBOR 6M grid; dynamics under the spot LIBOR measure:

$$\mathrm{d}F_i(t) = \sigma_i(t)\,\phi(F_i) \sum_{j=q(t)}^{i} \frac{\tau_j\,\rho_{ij}\,\sigma_j(t)\,\phi(F_j)}{1+\tau_j F_j(t)}\,\mathrm{d}t + \sigma_i(t)\,\phi(F_i)\,\mathrm{d}W_i(t). \tag{9}$$

[TODO] $\phi(F) = F + \delta$ (shifted lognormal) because of negative EUR rates.

7.2 Volatility and correlation parametrisation

[TODO] Rebonato's *abcd* form $\sigma_i(t) = (a + b(T_i - t))e^{-c(T_i-t)} + d$; correlation $\rho_{ij} = \rho_\infty + (1-\rho_\infty)e^{-\beta|T_i - T_j|}$; factor reduction by PCA of the correlation matrix.

7.3 Swap rates as baskets of forwards

[TODO] $S(t) = \sum_i w_i(t) F_i(t)$; Rebonato's frozen-weight approximation of the swaption volatility; this gives the calibration to swaptions and the correlation of CMS rates.

7.4 SABR for each forward

$$\mathrm{d}F = \alpha\,(F+\delta)^\beta\,\mathrm{d}W, \qquad \mathrm{d}\alpha = \nu\,\alpha\,\mathrm{d}Z, \qquad \mathrm{d}\langle W, Z\rangle = \rho\,\mathrm{d}t. \tag{10}$$

[TODO] Hagan's implied vol expansion (normal-vol version); shifted SABR; arbitrage at low strikes and its relevance for CMS replication.

7.5 SABR-LMM

[TODO] Rebonato–McKay–White or Mercurio–Morini version: one stochastic volatility per forward or a common volatility factor, correlation between forwards and volatilities, drift corrections. Choose one and justify.

7.6 Simulation and pricing of CMS rates

[TODO] Predictor–corrector or log-Euler on the forwards; CMS rates at each fixing are computed from the simulated forwards; cost, variance reduction (control variate with the replicated CMS rate).

7.7 Calibration

[TODO] Two steps: SABR per swaption expiry/tenor (smile), then LMM *abcd* and correlation to the ATM matrix and to the CMS correlation target.

7.8 Limitations

[TODO] High dimension, slow Monte Carlo, calibration instability, approximate swaption formulas.

8 Pricing the CMS Steepener

[**TODO**] State clearly the pricing date, the curve, and the origin of the volatility inputs (Remark 4.1).

8.1 Convexity adjustments

[**TODO**] Table: CMS 2y and 10y convexity adjustments by fixing date, from replication, Hagan's formula, G2++, Cheyette and SABR-LMM.

8.2 Correlation implied by each model

[**TODO**] Figure: terminal correlation of S_2 and S_{10} as a function of fixing date for each model, against the historical rolling correlation.

8.3 Steepener prices

Model	Price (% notional)	MC std. error	Run time
Bachelier spread (flat ρ)	–	–	–
G2++	–	–	–
Cheyette local vol	–	–	–
Cheyette stochastic vol	–	–	–
SABR-LMM	–	–	–

Table 1: [**TODO**] Fill in. Steepener of (2).

8.4 Sensitivities

[**TODO**] Delta to curve level and slope, vega, sensitivity to correlation, fair margin m or fair leverage L.

9 Comparison of the Models

[**TODO**] Table: factors, smile, correlation control, calibration speed, pricing speed, suitability for callables (Bermudan steepener). Discussion of which model a desk would use for which product.

10 Conclusion

[**TODO**] Write last.

A Proofs

[**TODO**] Measure change (3); replication formula (4); G2++ bond price; quasi-Gaussian bond reconstruction formula.

B Implementation

- `Fixed_Income_Models/sabr.py`: normal and shifted SABR smile (`SABR`).
- `Fixed_Income_Models/cms.py`: forward swap rates, linear-TSR static replication (`CMSReplication`), swap rates and swaption smiles from any simulation model.
- `Fixed_Income_Models/hull_white_2F.py`: G2++ (`G2pp`), exact simulation, frozen-gradient swaption vols and correlations, calibration.
- `Fixed_Income_Models/cheyette.py`: d-factor quasi-Gaussian model with local and stochastic volatility (`QuasiGaussian`), Euler Monte Carlo, smile calibration.
- `Fixed_Income_Models/lmm.py`: SABR-LMM (`SABRLMM`), spot-measure predictor–corrector simulation, Rebonato calibration.
- `Products/Steepener_CMS/eur_swap_market.py`: data, curve, historical statistics, volatility proxies.
- `Products/Steepener_CMS/steepener_cms.py`: the note, the replication + Gaussian spread pricer, the model-agnostic Monte Carlo pricer, the fair margin.
- `Products/Steepener_CMS/Steepener_CMS_pricer.ipynb`: every number and figure of Section 8 (figures saved in `Reports/CMS/figures/`).

[**TODO**] Code excerpts: CMS replication, G2++ step covariance, Cheyette Euler step, LMM drift.

C Notation

[**TODO**] Notation table.

References

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